

No Cortana for kids

Microsoft's newly released voice assistant won't talk to kids under 13 years due to child protection laws. More here: <http://dgit.in/CortaNo>



Drones catching wind

Saudi Arabia recently bought its first drone – the “Pterodactyl” for surveillance. Read more here: <http://dgit.in/ShotADrone>

LED Monitors

JARGON BUSTER

TN: Twisted Nematic (TN) panels are among the cheapest type of panels available. Without going into too much detail, these panels generally offer high refresh rates, an average contrast ratio and decent viewing angles with average colour reproduction.

VA: In these panels, the liquid crystals are perpendicular to the glass substrate and tilt to allow light to pass through. These panels generally offer great contrast, deep blacks and excellent colour reproduction. The refresh rate is usually low however, and colour reproduction is not as good as on IPS panels.

IPS: In-Plane-Switching, in this type of display technology, the liquid crystals are aligned parallel to the glass substrate and are re-oriented in the same plane to allow light to pass through. However, don't fret if this doesn't make sense. All you need to know is that IPS generally offers excellent viewing angles and possibly the best colour reproduction among LCD monitors. IPS panels do suffer from low refresh-rates and they tend to be large, expensive and generate a considerable amount of heat. That said, there are many variants of IPS displays, including e-IPS, s-IPS, PLS and many

more. Each offers some sort of advantage and compromise over the other.

Passive-Matrix LCD: This is a now defunct (thankfully) technology that was used in the early days of the LCD. Basically, a grid-pattern circuit would be responsible for the switching of the liquid crystals. A comparatively rudimentary process that was slow and was responsible for low refresh rates. It was cheap though.

Active-Matrix LCD: All modern LCD panels available today use this technology. In essence, each pixel is a transistor that can be turned on or off individually, vastly improving image quality and refresh rates because of the finer control.

Contrast Ratio: This is a measure of how much brighter a white scene is compared to a black one. Good monitors will have a high contrast ratio, but a high contrast ratio isn't everything. A contrast ratio of above 500:1 is perfectly adequate for most users.

Refresh Rate / Response Time: Refresh Rate is a measure of the frequency at which a monitor can “refresh” an entire image on its screen and is measured in Hz. Response Time is exactly the same thing, but it's measured

in ms. As a general rule of thumb, a higher refresh rate or lower response time is what you're looking for. Any monitor that says AMA or Flicker Free usually boasts of either a high refresh rate or a boost mode to pump up the refresh rate if required.

Backlight bleed: All LCD monitors incorporate a source of light, usually in the form of an LED array. The orientation of the liquid crystals on the monitor, controlled via some circuitry, limits the amount of light that escapes, generating lighter and darker shades. Backlight bleed is visible when the crystals do not adequately block the light emanating from the backlight. All monitors suffer from some degree of backlight bleed. How much is acceptable is a matter of personal preference.

Low Blue Light: Many of the BenQ monitors in the test came with a setting that said “Low Blue Light”. This mode is supposed to reduce the amount of blue light emitted by the backlight of a monitor, thereby reducing the stress on our eyes. Surprisingly, it actually worked and we found that for regular browsing and office work, the mode worked wonders. Give it a spin if you get the chance.

EA273WM-BK hadn't been around, this one would have walked away with the “Best Performer” crown yet again.

BEST PERFORMER NEC EA273WM-BK

We definitely didn't expect a TN panel to win our best performer award and definitely not one with a 1080p screen resolution on a 27-inch panel. However, this panel, despite its abysmally low brightness of 87cd/m2 (once calibrated), sports the lowest levels of bleed we've seen yet. Backlight bleeding does increase once you pump up the brightness, but since we're only measuring calibrated levels, this monitor is unmatched and even at higher brightness levels the contrast ratio still holds. This ensures

consistent performance under any circumstances.

Sporting a TN panel, this monitor does also boast of a lower response time than most other monitors in this category and that response time coupled with the low bleed means that movies and games look stunning. For still images you'd still want to use something like ASUS's PB278Q, but for everything else, this NEC panel is just better.

The monitor also features a sensor that dynamically adjusts brightness based on ambient light, but more interestingly, it can also check for the presence of a human in front of the screen and tones down the brightness considerably when one is absent. This feature is interesting, but was a bit irritating to use in practice

since you have to be directly in front of the monitor for this to work. If you adopt the correct posture at your desk you will always be directly in front of your monitor anyway, so there's that to consider.

Do note that this is a very expensive monitor but if you do purchase it, you won't regret your decision.

BEST BUY BenQ GL2760H

If it is value for money you want then the GL2760H is the monitor you'd want to buy. The monitor is strictly average in every regard, except the brightness value (300 cd/m2, which is very high), and performs reasonably well at everything. There is quite a bit of backlight bleed in dark scenes, but not much more than the other

monitors in this category and it's definitely within an acceptable range. We can't find anything particularly good about this monitor, but then again there's nothing particularly bad about it. What tips the balance in its favour is the price and adequate performance. This is the best 27-inch monitor that the average user can buy today.

This test was definitely an interesting one and one that we learnt a lot from. Hopefully, the detailed tables and buying guide should help you reach an intelligent decision with regards to the best monitor for your needs. If not, we do have a buying guide on page 92 if you don't want to go through all the statistics. We're always looking forward to hearing from you so do write in to us at: editor@thinkdigit.com